PROJECT SUMMARY

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Gaussian Mean Testing under Communication Constraints with s shared random bits

Introduction. In a distributed setting, users draw samples from a d dimensional spherical Gaussian distribution $\mathcal{G}(\mu, \mathbb{I}_d)$. A central referee seeks to distinguish between the two hypotheses: (i) $\|\mu\|_2 = 0$ or (ii) $\|\mu\|_2 \geq \varepsilon$ for some pre-specified threshold $\varepsilon \in (0,1)$ with high probability. While users have access to s shared random bits, they can transmit only ℓ bits to central referee. This is the distributed version of the classic gaussian mean testing problem. We describe a protocol for this problem with sample complexity, i.e., minimum number of users required to achieve the task, $O(d/\sqrt{l}\varepsilon^2 \cdot (l/\ell))$ where $l = (2d/b) \vee \ell$ and $b = 2^{\lfloor s/40 \rfloor} \wedge d$. For brevity, we call this problem distributed gaussian mean testing.

Previous Work. The problem was previously studied by Acharya et al. (2020) in the settings where users shared no randomness (*private coin*) and infinite randomness (to sample a rotation matrix u.a.r.) (*public coin*). They described the protocols for the two settings with sample complexity $O(d^{3/2}/\ell\varepsilon^2)$ in private coin and $O(d/\sqrt{\ell}\varepsilon^2)$ in public coin. They note that randomness is useful in the testing as private coin has factor $\sqrt{d/\ell}$ worse sample complexity that public coin. However, the setting when users have access to only finite shared randomness remains unstudied. We make significant progress towards understanding this regime by designing protocol for setting when users share s random bits.

Our results. We describe an s-coin protocol for distributed gaussian mean testing, where users have access to s shared random bits. Theorem 1 describes the sample complexity of the protocol. Note that $s = 0$ is the *private coin* setting, for which, Theorem 1 shows sample complexity $O(d^{3/2}/\ell\varepsilon^2)$ which matches with results of Acharya et al. (2020). With $s = O(\log(d/\ell)$, Theorem 1 gives sample complexity $O(d/\sqrt{\ell}\varepsilon^2)$, thus, improving randomness requirement for *public coin* from infinity to $O(\log(d/\ell))$.

Theorem 1 (Distributed Gaussian Mean Testing, *s-coin*)**.** *There exists a protocol for distributed gaussian mean testing under ℓ-bit communication with $O\left(\frac{d}{\sqrt{l}\varepsilon^2} \cdot (\frac{l}{\ell})\right)$ users sharing s random bits where $l = (2d/b) \vee \ell$ and $b = (2^{\lfloor s/40 \rfloor}) \wedge d$.*

References

Jayadev Acharya, Clément L Canonne, and Himanshu Tyagi. Distributed signal detection under communication constraints. In Jacob Abernethy and Shivani Agarwal, editors, *Proceedings of Thirty Third Conference on Learning Theory*, volume 125 of *Proceedings of Machine Learning Research*, pages 41–63. PMLR, 09–12 Jul 2020. URL https://proceedings.mlr.press/v125/acharya20b.html.